Functional Equations

Version B

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December 11, 2018

§1 Introduction

Recall trivia:

Definition 1.1. A function $f : A \mapsto B$ is a rule, that assigns to each $x \in A$ a unique element $f(x) \in B$.

Think of a function as a collection of *arrows* emanating from each element of set A (domain), with each arrow pointing at exactly one element of B (range).

Example 1.2

Some functions are as given below:

- $f : A \mapsto A$, $f(x) = x$ for all $x \in A$.
- All polynomials, exponentials, logarithmic functions blah blah.
- $f : \mathbb{R} \mapsto [-1, 1]$, $f(x) = \sin x$.
- $f : \mathbb{R} \mapsto \{0, 1\}$, where $f(x) = 0 \iff x \in \mathbb{Q}$.

The last example tells you that functions could be *arbitrarily* bad.

So the idea behind "solving a functional equation", say

Example 1.3

$f : \mathbb{Q} \mapsto \mathbb{R}$ with $f(x + y) = f(x) + f(y)$ for all x, y

is that, well, I give you this *huge* system of equations:

$$f(1.7) = f(1) + f(0.7), f(2.5) = f(1) + f(1.5), f(3.2) = f(2) + f(1.2), \ldots$$

and ask you, "a characterization for the collection of all such functions". As you may have guessed (and what we'll talk about in detail later), the solution set consists of all functions $x \mapsto cx$, where $c \in \mathbb{R}$ is arbitrary.

Remark 1.4. The method of proof is "two-ways" implication. First, you assume that f satisfies the hypothesis and then deduce that it must lie in the "claimed" solution set. Then, you need to check that each member of the solution set is a valid function. Failure to do either makes a solution incomplete!

In fact, I recommend this as the opening line of your solution:

Proof. We claim that the only solutions are functions of the form $x \mapsto cx$, where $c \in \mathbb{R}$ is arbitrary. First, we check that these are in fact solutions. This is clear.

Now we prove that these are the only such functions. Indeed, let f satisfy... then... $\square$

Finally, some more useful definitions:

Definition 1.5. A function $f : A \mapsto B$ is called *injective* if $f(x) = f(y) \iff x = y$.

Definition 1.6. A function $f : A \mapsto B$ is called *surjective* if for each $y \in B$, there exists $x \in A$ such that $f(x) = y$.

Definition 1.7. A function $f : A \mapsto B$ is called *bijective* if f is both injective and surjective.

Definition 1.8. A function $f : A \mapsto A$ is called an *involution* if $f(f(x)) = x$ for all $x \in A$.

Lemma 1.9
All involutions are bijections.

§2 Tropes and Examples

§2.1 Point-wise Trap

Example 2.1
Find all functions $f : \mathbb{R} \mapsto \mathbb{R}$ such that $f(x)^2 = 1$ for all $x \in \mathbb{R}$.

A common pitfall is to take square roots and say $f(x) = 1$ and $f(x) = -1$ are the two solutions. No, in fact, for any disjoint union of $\mathbb{R}$, we can assign all elements of one set as $+1$ and all elements of the other as -1. Check that these are all possible solutions.

Finally, some real contest examples.

Example 2.2 (INMO 2015)
Let $\mathbb{R}$ denote the set of all real numbers. Find all real functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x^2 + yf(x)) = xf(x + y)$$

for all $x, y \in \mathbb{R}$.

Example 2.3 (IMO 2008/4)
Find all functions $f : (0, \infty) \mapsto (0, \infty)$ (so f is a function from the positive real numbers) such that

$$\frac{(f(w))^2 + (f(x))^2}{f(y^2) + f(z^2)} = \frac{w^2 + x^2}{y^2 + z^2}$$

for all positive real numbers w, x, y, z, satisfying $wx = yz$.

As bonus, try this one:

Example 2.4 (USAMO 2016/4)
Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that for all real numbers x and y,

$$(f(x) + xy) \cdot f(x - 3y) + (f(y) + xy) \cdot f(3x - y) = (f(x + y))^2.$$

§2.2 Cauchy's Functional Equation

We revisit the baby example we started with early on.

Example 2.5
Find all solutions to $f : \mathbb{Q} \mapsto \mathbb{R}$ such that $f(x + y) = f(x) + f(y)$ for all x, y.

Proof. Plug iteratively to get $f(n) = nf(1)$ for all $n \in \mathbb{N}$. Get $f(0) = 0$ and f odd. Then write $f\left(\frac{p}{q}\right) = pf\left(\frac{1}{q}\right)$ and $f(1) = qf\left(\frac{1}{q}\right)$ to conclude. □

Now, the real problem is that this equation when $f : \mathbb{R} \mapsto \mathbb{R}$ has **terribly** many bad solutions. We won't bother characterizing them all. We will just state, without proof, the following useful result:

Lemma 2.6 (Cauchy with conditions implies linear)
Let $f : \mathbb{R} \mapsto \mathbb{R}$ be an additive function. Then any of the following implies f is linear.

- f is continuous.
- f is bounded on an interval.
- f is monotonic.

In particular, all field automorphisms of $\mathbb{R}$ are identity.

Proof. We prove the automorphism part. □

The idea is to try and reduce a given problem into Cauchy's equation with some conditions and conclude.

Example 2.7 (India TST 2016/2)
Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f\left(x^3 + f(y)\right) = x^2 f(x) + y,$$

for all $x, y \in \mathbb{R}$.

§2.3 Symmetrization

The idea is to switch variables in an unsymmetric condition to gather more information.

Example 2.8 (Singapore 1999)

Find all functions $f : \mathbb{R} \mapsto \mathbb{R}$ such that

$$(x-y)f(x+y)-(x+y)f(x-y)=4xy(x^2-y^2).$$

Example 2.9 (IMO ShortList 2005 A2)

We denote by $\mathbb{R}^+$ the set of all positive real numbers.

Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ which have the property:

$$f(x)f(y)=2f(x+yf(x))$$

for all positive real numbers x and y.

Remark on substitutions: Two common ideas are to (a) force cancellations and (b) to make clever substitutions. Use information like injectivity/surjectivity to get more substitutions. It's not much use to give an explicit example, so we'll use it implicitly everywhere else!

§2.4 Induction and Discrete tricks

For FEs with $\mathbb{N}$ as the domain, it is often useful to try induction, well-ordering or other discrete ideas. We'll do some examples here.

Example 2.10 (IMO 2009/5)

Determine all functions f from the set of positive integers to the set of positive integers such that, for all positive integers a and b, there exists a non-degenerate triangle with sides of lengths

$$a, f(b) \text{ and } f(b+f(a)-1).$$

Example 2.11 (EGMO 2017/2)

Find the smallest positive integer k for which there exists a colouring of the positive integers $\mathbb{Z}_{>0}$ with k colours and a function $f : \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ with the following two properties:

(i) For all positive integers m, n of the same colour, $f(m+n)=f(m)+f(n)$.

(ii) There are positive integers m, n such that $f(m+n) \neq f(m)+f(n)$.

In a colouring of $\mathbb{Z}_{>0}$ with k colours, every integer is coloured in exactly one of the k colours. In both (i) and (ii) the positive integers m, n are not necessarily distinct.

Example 2.12 (IMO 1997/6)

Find all functions $f : \mathbb{N} \mapsto \mathbb{N}$ such that $f(n+1) > f(f(n))$ for all $n \in \mathbb{N}$.

Example 2.13 (IMO ShortList 2015 A2)
Determine all functions $f : \mathbb{Z} \to \mathbb{Z}$ with the property that

$$f(x - f(y)) = f(f(x)) - f(y) - 1$$

holds for all $x, y \in \mathbb{Z}$.

§3 Practice Problems

Miscellaneous problems for practice.

§3.1 Easy

Problem 3.1 (INMO 2011/6)**.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying

$$f(x+y)f(x-y) = (f(x) + f(y))^2 - 4x^2 f(y),$$

For all $x, y \in \mathbb{R}$.

Problem 3.2 (IMO 2010/1)**.** Find all function $f : \mathbb{R} \to \mathbb{R}$ such that for all $x, y \in \mathbb{R}$ the following equality holds

$$f(\lfloor x \rfloor y) = f(x) \lfloor f(y) \rfloor$$

where $\lfloor a \rfloor$ is greatest integer not greater than a.

Problem 3.3 (USAMO 2002/4)**.** Find all functions $f : \mathbb{R} \mapsto \mathbb{R}$ satisfying

$$f(x^2 - y^2) = xf(x) - yf(y)$$

for all x, y.

Problem 3.4 (Iran TST 1996)**.** Find all functions $f : \mathbb{R} \mapsto \mathbb{R}$ satisfying

$$f(x^2 + y) = f(f(x) - y) + 4f(x)y$$

for all x, y.

Problem 3.5 (TSTST 2018/1)**.** As usual, let $\mathbb{Z}[x]$ denote the set of single-variable polynomials in x with integer coefficients. Find all functions $\theta : \mathbb{Z}[x] \to \mathbb{Z}$ such that for any polynomials $p, q \in \mathbb{Z}[x]$, $\theta(p+1) = \theta(p) + 1$, and if $\theta(p) \neq 0$ then $\theta(p)$ divides $\theta(p \cdot q)$.

§3.2 Medium

Problem 3.6 (IMO 2012/4)**.** Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that, for all integers a, b, c that satisfy $a + b + c = 0$, the following equality holds:

$$f(a)^2 + f(b)^2 + f(c)^2 = 2f(a)f(b) + 2f(b)f(c) + 2f(c)f(a).$$

(Here $\mathbb{Z}$ denotes the set of integers.)

Problem 3.7 (ELMO ShortList 2013 A3)**.** Find all $f : \mathbb{R} \to \mathbb{R}$ such that for all $x, y \in \mathbb{R}$, $f(x) + f(y) = f(x + y)$ and $f(x^{2013}) = f(x)^{2013}$.

Problem 3.8 (India TST 2018/3)**.** Find all functions $f : \mathbb{R} \mapsto \mathbb{R}$ such that

$$f(x)f(yf(x) - 1) = x^2 f(y) - f(x),$$

for all $x, y \in \mathbb{R}$.

Problem 3.9 (IMO ShortList 2005 A4)**.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x + y) + f(x)f(y) = f(xy) + 2xy + 1$ for all real numbers x and y.

§3.3 Hard

Problem 3.10 (IMO ShortList 2016 A4)**.** Find all functions $f : (0, \infty) \to (0, \infty)$ such that for any $x, y \in (0, \infty)$,

$$xf(x^2)f(f(y)) + f(yf(x)) = f(xy)\left(f(f(x^2)) + f(f(y^2))\right).$$

Problem 3.11 (IMO ShortList 2009 A7)**.** Find all functions f from the set of real numbers into the set of real numbers which satisfy for all x, y the identity

$$f\left(xf(x+y)\right) = f\left(yf(x)\right) + x^2$$

Problem 3.12 (IMO 2013/5)**.** Let $\mathbb{Q}_{>0}$ be the set of all positive rational numbers. Let $f : \mathbb{Q}_{>0} \to \mathbb{R}$ be a function satisfying the following three conditions:
(i) for all $x, y \in \mathbb{Q}_{>0}$, we have $f(x)f(y) \geq f(xy)$;
(ii) for all $x, y \in \mathbb{Q}_{>0}$, we have $f(x+y) \geq f(x) + f(y)$;
(iii) there exists a rational number $a > 1$ such that $f(a) = a$.
Prove that $f(x) = x$ for all $x \in \mathbb{Q}_{>0}$.

Problem 3.13 (IMO 2015/5)**.** Let $\mathbb{R}$ be the set of real numbers. Determine all functions $f : \mathbb{R} \to \mathbb{R}$ that satisfy the equation

$$f(x + f(x+y)) + f(xy) = x + f(x+y) + yf(x)$$

for all real numbers x and y.

Problem 3.14 (APMO 2010/5)**.** Find all functions f from the set $\mathbb{R}$ of real numbers into $\mathbb{R}$ which satisfy for all $x, y, z \in \mathbb{R}$ the identity

$$f(f(x) + f(y) + f(z)) = f(f(x) - f(y)) + f(2xy + f(z)) + 2f(xz - yz).$$